

Lecture Unit 3.6

Stability

Stability

- We need to consider the **stability** of the numerical solution as well as the **accuracy**
- The solution must be guaranteed to remain finite for the scheme to be stable
- If errors in the solution can grow without bound the numerical scheme is unstable
- Carrying out a stability analysis is beyond the scope of this module but we will consider the factors which affect stability and types of stability

Factors affecting stability

- The stability of a numerical method depends on:
 - the equation being solved
 - how the spatial derivatives are calculated
 - how the time steps are calculated

Types of stability

- A numerical method may be:
 - **Unconditionally stable:** errors are damped and the solution is always stable
 - **Unconditionally unstable:** errors are amplified and the solution is always unstable
 - **Conditionally stable:** solution is stable subject to a constraint

Forward in time centred in space

- The time step used in the previous examples (units 3.2-3.4) is called a “forward-Euler” time step
- If we combine this with a centred difference for the spatial derivatives

$$\frac{\partial u}{\partial x} \approx \frac{u_{i+1} - u_{i-1}}{2\Delta x}$$

and apply it to the advection equation the resulting scheme is unconditionally unstable 😞

Forward in time with one-sided difference

- If we use a forward-Euler time step with one-sided spatial difference

$$\frac{\partial u}{\partial x} \approx \frac{u_i - u_{i-1}}{dx} \quad \text{or} \quad \frac{\partial u}{\partial x} \approx \frac{u_{i+1} - u_i}{dx}$$

to solve the advection equation the resulting scheme is stable provided

$$\Delta t \leq C \frac{\Delta x}{v_{\max}}$$

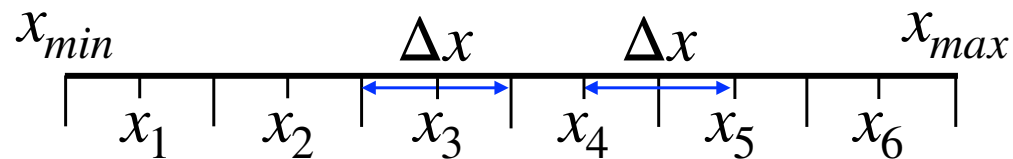
where $0 < C < 1$ is a constant (Courant number) and $v_{\max}$ is the maximum velocity

- This is a **conditionally stable** scheme and the stability condition is known as the CFL (Courant–Friedrichs–Lewy) condition

CFL condition

$$\Delta t = C \frac{\Delta x}{v_{\max}} \implies \Delta x = \frac{\Delta t v_{\max}}{C}$$

$C < 1$ means that no material can advect further than Δx in time Δt



Unit Summary

- The solution must remain finite for a numerical scheme to be stable
- Stability depends on the equation solved, time stepping and how spatial gradients are calculated
- A numerical scheme can be unconditionally stable, unconditionally unstable or conditionally stable
- A forward-Euler time step and centred difference is unconditionally unstable for the advection equation
- A forward-Euler time step and one-sided difference is conditionally stable for the advection equation: it is stable for $\Delta t \leq C \frac{\Delta x}{v_{\max}}$